Network-Constrained P2P Trading: A Safety-Aware Decentralized Multi-Agent Reinforcement Learning Approach

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***Abstract*—With increasingly predominant renewable energy sources and market deregulation in the distribution network, considerable research effort has been devoted to safe reinforcement learning (RL). Network-constrained peer-to-peer (P2P) transaction is particularly challenging, as the system-wide global constraint is implicitly determined by the joint behavior of individual agents. Existing safe RL methods are burdened of addressing this type of constraint. In this context, this paper proposes a decentralized Multi-Agent Safety-Aware Learning (MASAL) method based on trust region concept. It treats system operation safety as a prerequisite and realizes safety-aware target. Furthermore, a system-wide safety condition classification is proposed, categorizing trading policy optimization into five operation conditions. Endogenous and exogenous violations can be distinguished. Building on the classification of system operation conditions, the method conducts preventive trading policy under normal conditions, and responsive policy under critical conditions when exogenous violations occur. Case studies show that the proposed method outperforms state-of-the-art safety layer method and Lagrangian relaxation method under normal and critical conditions. Numerical results also highlight the effectiveness of the method in terms of interpretable trading policies, scalability performance, and training robustness with multiple constraints.**

***Index Terms*—Safe multi-agent reinforcement learning, trust region methods, peer-to-peer transaction, system-wide constraint.**

I. Introduction

PEER-TO-PEER (P2P) energy trading in distribution networks (DN) has three features: 1) The traditional centralized DN operation is transitioning to a distributed and cooperative approach [1], so the system-wide constraints are determined by the joint behavior of individual operators. 2) Given the short market clearing interval of five or fifteen minutes, the system operation conditions are varied and highly dynamic. 3) P2P Transaction introduces multiple participants in network operation. These features pose challenges to coordinate participants' compliance of the physical DN constraints and their economic objective at the same time [2]. It is critical to implement P2P trading without compromising network security [3].

Conventionally, optimization-based methods employed in network-constrained P2P can be broadly divided into three types: 1) Centralized approach [4] assumes the DN and all of the P2P participants are managed by a centralized coordinator. 2) Bi-level negotiation approach [5] executes P2P transaction at the lower level to obtain the equivalent load. The trading adjustment and DN operation management are conducted at the upper level. 3) Consensus-based distributed approach [6], [7] decomposes the global optimal power flow problem into local subproblems through distributed algorithms, such as alternating direction method of multipliers. Apart from the optimization-based methods, [8] employs a heuristic method for the household-level P2P trading in a low-voltage network, but it is not applicable to medium-voltage distribution networks. However, limited by the computational burden of multiple iterations, conventional methods struggle to fully capture the P2P market dynamics.

In this context, intelligent and autonomous decision-making methods is urgently demanded. Specifically, deep reinforcement learning (DRL) has been considered an potential approach [9]. RL training is based on trial-and-error and interaction with an environment. It enables close-to-real-time decision-making in P2P market (typically on time scales of milliseconds), demonstrating fast response speed to dynamic operation conditions [10], [11].

However, the vanilla DRL training relies on random exploration and it formalizes an unconstrained Markov Decision Process (MDP). It inevitably generates actions that violate vital physical network constraints and may cause catastrophic outcomes, such as P2P trading without network safety concern in [12]. It triggers considerable research efforts devoted to the safe RL for modern power systems [13]. Safe RL learns the policies that maximize reward and adheres to the safety constraints [14]. Based on whether they are coupled with the RL framework, existing safe RL methods can be

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coarsely categorized into two types [15]: Safety layer methods [16], [17] and safe policy optimization methods [18], [19], [20], [21], [22], [23], [24], [25], [26], [27].

Safety layer methods in power system add a safety layer to rectify unsafe actions as a check before interacting with the environment, which are decoupling from the conventional RL framework. One way to modify unsafe actions is *action projection*. In [16], a quadratic programming (QP) layer correcting the primitive action is embedded in the actor neural network, which keeps the voltage magnitude within the safe range. Reference [17] employs appliance-specific logic to deal with local constraints in building energy management (BEM), which are determined by individual agents such as heating ventilation, and air conditioning (HVAC) and washing machine.

Safety layer methods are usually used for constraints that the safety cost value is a linear function dependent on the action. Local constraints, such as state of charge of energy storage (ES) corresponding to charging power, and room temperature corresponding to HVAC operation setting point, are typical examples. However, they have low adaptability in dynamic and intricate network-constrained P2P trading environments. The pre-trained safety examination process evaluating whether the original action is safe relies heavily on the parameterized system model. It lacks universality and cannot be adaptive to various DN operation conditions.

Coupled with the traditional RL framework, *safe policy optimization methods* deal with constraints by transforming policy optimization criterion. They formulate Constrained Markov Decision Process (CMDP) instead of conventional MDP to explicitly describe the constraint violation. To solve the CMDP, one popular method is *reward shaping*. It introduces an auxiliary penalty component in reward function and reformulates an unconstrained problem. A direct way of reward shaping is penalizing violation with a fixed penalty factor [18], [19], which necessitates manually tuning penalty factors and expert knowledge. As an extension of reward shaping, *Lagrangian relaxation method* is commonly used to formulate penalty term. It balances the trade-off between reward and safety cost [20], [21], [22], [23].

The Lagrangian relaxation method is applicable to constraints that the safety cost value is explicitly dependent on the action, such as local voltage-reactive power [24]. However, this method merely combines safety and non-safety performance metrics into the reward function, but the exact gradient of safety violation cost is overlooked.

Notably, the system-wide network constraint in P2P transaction is indirectly and implicitly impacted by the actions (bidding of agents), which cannot be expressed as a function of the actions. The nodal voltages are affected by decentralized bidding actions which intend to pursue high trading profits. To tackle the complex constraint, there is an urgent need to utilize the safety cost gradient in guiding safe policy search.

Furthermore, constraint violations can be categorized into two types based on their causes in a P2P energy trading problem: 1) Endogenous violations arising from transactions. For example, a participant may increase the local generation to sell energy, causing a large net injection that leads to voltage exceeding the upper limit. 2) Exogenous violations resulting from the DN, regardless of the trading intentions. For instance, fluctuations in output from a utility-owned renewable energy station due to weather conditions can cause the voltage violation. To earn profits within a secure distribution system, it is pressingly needed for participants to distinguish normal conditions when violations stem from endogenous transactions, and critical conditions when violations stem from exogenous DN.

TABLE I
COMPARISONS OF SAFE RL IN EXISTING LITERATURE

Ref	Method	Single/ Multi-agent	Problem	Constraint indirectly affected by actions
[16]	SL	S	VVC	-
[17]	SL	M	BEM	-
[18]	penalty	M	VVC	-
[19]	Consensus+penalty	M	P2P transaction	√
[21],[22]	Lag	S	EV management	-
[23]	Lag	S	OPF	-
[25]	Consensus+TR	M	Networked MGs management	-
[26]	TR	S	DN operation	-
[27]	TR	S	EVCS management	-
This paper	TR	M	P2P transaction	√

SL: safety layer Lag: Lagrangian relaxation, TR: trust region, S: single, M: multi

Different from striking a balance between safety and non-safety performance, another popular approach is the *trust region method*. It directly addresses the constrained optimization problem. It separately treats the economic objective and safety constraints. Reference [25] develops a trust-region-based multi-agent safe RL method for networked microgrids (MGs), but it requires costly inter-agent communication to achieve the consensus. As a representative, constrained policy optimization (CPO) algorithm is employed to optimize DN operation [26], [27]. Although this method succeeds in the context of single-agent systems, the coordination of trading policies is challenging when extended to decentralized multi-agent systems owing to conflicts among agents [28]. Detailed comparisons of safe RL methods are listed in Table I.

To the best of our knowledge, although research effort has been devoted to address the constraints explicitly determined by actions, no safe RL method is developed for the complicated network constraint of P2P transaction. Most safe RL methods are single-agent, which does not align with the autonomous nature of decentralized market participants. Therefore, this paper proposes a multi-agent deep reinforcement learning (MADRL) method to deal with system-wide global constraint indirectly impacted by joint trading action. The key contributions are threefold:

1) A decentralized multi-agent safe policy learning method based on trust region concept is proposed. It addresses the network-constrained P2P transaction of multiple microgrids (MMGs) in repeated double auction (RDA) market. It

eliminates conflicts among agents and agents can reach a consensus towards safe joint trading policy.

2) A system-wide safety condition classification is proposed, categorizing trading policy optimization into five operation conditions. Building on the classification, safety-aware policy update scheme is developed, which conducts preventive trading policy under normal conditions when endogenous violations occur, and responsive policy under critical conditions when exogenous violations occur.

3) Unlike striking a balance between safety and non-safety performance through weight coefficients under Lagrangian relaxation methods, safety is treated as a prerequisite constraint, rather than merely an aspect of policy performance. The safety cost gradient is explicitly integrated in the policy update direction, guiding the safe P2P trading policy search.

The remainder of this paper is organized as follows. Section II formulates the network-constrained P2P trading problem into a Constrained Cooperative Markov Game. Section III introduces the proposed MASAL method and the system-wide safety condition classification. In Section IV, numerical tests are performed. Finally, Section V draws the conclusions and identifies future directions.

II. Network-Constrained P2P Energy Trading in Repeated Double Auction Market

We concentrate on a network-constrained P2P energy trading problem of Distribution System Operator (DSO)-owned MMGs in RDA market [29], [30]. The market horizon is 24 hours with an hourly time step. Specifically, the MG is comprised of the renewable energy source (RES), energy storage (ES), micro turbine (MT) and demand response (DR), linked to a specific bus known as the Point of Common Coupling (PCC). Each MG can utilize an energy management system to schedule its flexibility resources. The net load injection of the PCC is determined by the MG operator. DSO, which is responsible for secure network operation, runs the power flow to calculate the voltage magnitude. After observing 1) the physical information of voltage and local demand and supply; 2) the market information of network tariff and clearing outcomes, the MG adjusts its price-quantity strategy. Each MG has three ways to supply its load: from local energy sources, from the other MGs and from the main grid. It bids as a buyer or seller with price and quantity at each time step. Buyer and seller MGs are paired based on RDA mechanism. Then the matched order quantities are cleared at the matched price, while unmatched portions are balanced through transactions with the upstream grid. All formulas in this paper hold for ∀t.

A. Repeated Double Auction Mechanism

To clear the pairwise trading quantities and prices, and achieve the social welfare maximization, the RDA market clearing mechanism is introduced. The RDA mechanism operates within predetermined intervals known as *auction period* (1h) during the trading day. After submitting their order information at the beginning of each auction period, MGs are divided into a set of buyers $\mathcal{B} = \{1, \ldots, N_b\}$ and sellers $S = \{1, \ldots, N_s\}$. A buyer is a MG with energy deficit $q_{b,t}$ and is ready to purchase energy at the bid price $c_{b,t}$. A seller is a MG with energy surplus $q_{s,t}$ and is willing to sell energy at the ask price $c_{s,t}$. To incentivize MGs to trade with peers, the bid and ask prices are bounded between the network tariff, as formulated in equation (1).

$$\lambda_t^{\mathrm{FiT}} \le c_{s,t}, c_{b,t} \le \lambda_t^{\mathrm{ToU}}, s \in S, b \in \mathcal{B} \tag{1}$$

where λ_t^{FiT} and λ_t^{ToU} are FiT and ToU price, respectively. Then, an *auctioneer* (DSO) collects the submitted information of MMGs and records them in the buyer and seller order books, k_t^b and k_t^s. According to the principle of price first, the buyer/seller order book is sorted in a descending/ascending order, as indicated in equations (2) and (3).

$$k_t^b = \left\{ \left(c_{b,t}, q_{b,t}\right) | c_{1,t} \ge \cdots \ge c_{b,t} \ge \cdots \ge c_{N_b,t}, b \in B \right\} \tag{2}$$
$$k_t^s = \left\{ \left(c_{s,t}, q_{s,t}\right) | c_{1,t} \le \cdots \le c_{s,t} \le \cdots \le c_{N_S,t}, s \in S \right\} \tag{3}$$

Based on the buyer and seller order books, the auctioneer plots the bidding and asking curves to find the intersection point. When the bid price exceeds the ask price (S^{mat} sellers and B^{mat} buyers to the left of the intersection point), trading pairs will be matched. Subsequently, the trading energy and prices are calculated pairwise along the price curves, which are given by equations (4) and (5).

$$p_{bs,t} = p_{sb,t} = min\left\{q_{b,t}, q_{s,t}\right\}, s \in S^{mat}, b \in B^{mat} \tag{4}$$
$$\lambda_{bs,t} = \lambda_{sb,t} = \left(c_{b,t} + c_{s,t}\right)/2, s \in S^{mat}, b \in B^{mat} \tag{5}$$

The unmatched quantities (S^u sellers and B^u buyers to the right of the intersection point) will be resolved by trading at the network tariff. Thus, it is assured that any imbalance of MMGs is completely cleared. After clearing, the current auction period closes, and MGs reset the order information for the next period. The above process will repeat itself until the trading day ends.

B. Energy Management and Bidding Model of MMGs

The objective function $H_{m,t}$ of MG m consists of three parts: 1) Negative utility function $C_{m,t}^U$; 2) P2P trading cost with other MGs $C_{m,t}^{P2P}$; 3) Energy trading cost with the main grid $C_{m,t}^{\mathrm{grid}}$.

$$\min_{\mathbf{u}} H_{m,t} = C_{m,t}^U + C_{m,t}^{\mathrm{grid}} + C_{m,t}^{P2P} \tag{6a}$$
$$C_{m,t}^U = \alpha_m^{\mathrm{dis}} \left(p_{m,t}^L - p_{m,t}^{\mathrm{acL}}\right)^2 + \alpha_m^{\mathrm{MT}} \left(p_{m,t}^{\mathrm{MT}}\right)^2 + \beta_m^{\mathrm{MT}} p_{m,t}^{\mathrm{MT}} \tag{6b}$$
$$C_{m,t}^{P2P} = \sum_{\iota}^{L} \lambda_{\iota,t}^M p_{\iota,t}^M \cdot \nVdash_{m,t} \tag{6c}$$
$$C_{m,t}^{\mathrm{grid}} = \lambda_t^{\mathrm{FiT}} \left[p_{m,t}^{\mathrm{grid}}\right]^- + \lambda_t^{\mathrm{ToU}} \left[p_{m,t}^{\mathrm{grid}}\right]^+ \tag{6d}$$

where $\mathbf{u}$ is the vector of decision variables, including the actual power consumption $p_{m,t}^{\mathrm{acL}}$, the power production of MT $p_{m,t}^{\mathrm{MT}}$, the charging/discharging power of ES $p_{m,t}^{ch}/p_{m,t}^{\mathrm{dis}}$ and the bidding price $c_{m,t}$. The load of MG includes elastic load and inelastic load [31]. The elastic load can be curtailed to earn more profits. $p_{m,t}^L$ is the desired demand and α_m^{dis} is the discomfort cost coefficient. Similar to [32], the DR discomfort cost is a quadratic function.α_m^{MT} and β_m^{MT} are the cost function parameters. For the P2P trading cost, ι is the index of matched peers. L denotes the corresponding maximum number of matched pairs. $p_{\iota,t}^{M}$ is the cleared trading quantity, and $\lambda_{\iota,t}^{M}$ is the clearing price with respect to the quantity. Total bidding quantity of a MG could be matched with multiple peers in a time interval. The indicator $\Bbbk_{m,t} = 1$ if $m \in B^{\text{mat}}$ and $\Bbbk_{m,t} = -1$ if $m \in S^{\text{mat}}$. For the retail energy cost, $p_{m,t}^{\text{grid}}$ is the energy trading with the main grid, where $[\,\cdot\,]^{+/-}$ denotes $max/min\{\cdot, 0\}$. $p_{m,t}^{M}$ and $p_{m,t}^{\text{grid}}$ are determined through the RDA-based market clearing process.

The operation constraints of MT and curtailable load are presented by (6e) and (6f), respectively.

$$0 \le p_{m,t}^{\text{MT}} \le \overline{p}_m^{\text{MT}} \tag{6e}$$

$$\varepsilon_m^{l} p_{m,t}^{L} \le p_{m,t}^{\text{acL}} \le p_{m,t}^{L} \tag{6f}$$

where $\overline{p}_m^{\text{MT}}$ represents the upper limit of MT generation, and the coefficient ε_m^{l} denotes the ratio of inelastic load.

The operation constraints of ES m are outlined as follows.

$$0 \le p_{m,t}^{ch} \le \overline{p}_m^{\text{es}} \tag{6g}$$

$$-\overline{p}_m^{\text{es}} \le p_{m,t}^{\text{dis}} \le 0 \tag{6h}$$

$$E_{m,t+1}^{\text{es}} = E_{m,t}^{\text{es}} + P_{m,t}^{ch}\Delta t\eta_m^{ch} + P_{m,t}^{\text{dis}}\Delta t/\eta_m^{\text{dis}} \tag{6i}$$

$${}_{m}^{\text{es}} \le E_{m,t}^{\text{es}} \le \overline{E}_m^{\text{es}} \tag{6j}$$

where $\overline{p}_m^{\text{es}}$ denotes the active power limit of the ES. Constraints (6g) and (6h) limits the maximum charging and discharging power. The state-of-charge (SoC) transition between t and $t+1$ is described in (6i), where $\eta_m^{ch}/\eta_m^{\text{dis}}$ indicates the charging/discharging efficiency. The SoC level $E_{m,t}^{\text{es}}$ is restricted within the range of energy capacity $[{}_{m}^{\text{es}},\ \overline{E}_m^{\text{es}}]$, as given in constraint (6j).

The bidding quantity $p_{i,t}^{\text{bid}}$ should be equal to the difference between local demand and generation as in Eq. (6k).

$$p_{m,t}^{\text{bid}} = p_{m,t}^{\text{acL}} - p_{m,t}^{\text{RES}} - p_{m,t}^{\text{MT}} - p_{m,t}^{\text{dis}} + p_{m,t}^{ch} \tag{6k}$$

where $p_{m,t}^{\text{RES}}$ denotes the RES power generation.

C. Constraints of Distribution Network

The DN can be described as an undirected graph $\mathcal{G}(W, \mathcal{E})$, where W and $\mathcal{E}$ are the collections of nodes and branches, respectively. $i, j \in W$ are index of nodes and ij$\in \mathcal{E}$ are index of branches. The secure operation of DN is governed by constraints (7)–(10):

$$P_{i,t} = \sum_{j\in W} |V_{i,t}||V_{j,t}|\big(G_{\text{ij}} \cos\theta_{ij,t} + B_{\text{ij}} \sin\theta_{ij,t}\big), \forall i \tag{7}$$

$$Q_{i,t} = \sum_{j\in W} |V_{i,t}||V_{j,t}|\big(G_{\text{ij}} \sin\theta_{ij,t} - B_{\text{ij}} \cos\theta_{ij,t}\big), \forall i \tag{8}$$

$$P_{i,t} = p_{i(m),t}^{\text{bid}} - P_{i,t}^{\text{load}}, \forall i, m \in \Omega^{\text{PCC}} \tag{9}$$

where $P_{i,t}$, $Q_{i,t}$ represent the active and reactive power injected into node i, respectively; $V_{i,t}$ is the voltage magnitude at node i; G_{ij} and B_{ij} signify the conductance and susceptance of branch from node i to node j; and $\theta_{ij,t}$ indicates the voltage phase difference between nodes i and j. Equation (9) illustrates the net injection is difference between the bidding quantity of MG $p_{i(m),t}^{\text{bid}}$ and the load of DN $P_{i,t}^{\text{load}}$, where $p_{i(m),t}^{\text{bid}}$ is obtained by the MG solving problem (6). As given in Eq. (6k), $p_{i(m),t}^{\text{bid}}$ is determined by the active power of the ES, MT generation, RES output, and actual load consumption. The bus voltage magnitudes are limited by constraint (10).

$$V_i^{-} \le V_{i,t} \le V_i^{+}, \forall i \tag{10}$$

where $[V_i^{-}, V_i^{+}]$ is the safe range of voltage.

D. Constrained Cooperative Markov Game Formulation

Since the current bidding strategy of MMGs is influenced by the market clearing outcomes of the previous auction period, the decision-making of MGs apparently exhibits the Markov Property. Thus, we consider the P2P trading of MMGs in RDA-based market as a decentralized Constrained Cooperative Markov Game with 24-hour intervals. It minimizes the MMGs' joint energy costs while satisfying the system-wide voltage constraint.

Each MG acts as an agent responsible for orchestrating its energy scheduling and bidding strategy. The RDA-based P2P market is assumed as the environment. The Constrained Cooperative Markov Game is defined by a tuple $\langle \mathcal{M}, \mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{T}, R, C, \gamma\rangle$ where $\mathcal{M}$ agents with a set of state $\mathcal{S}$, a collection of observations $\mathcal{O}$, a set of joint actions $\mathcal{A} = A_1 \times A_2 \times \cdots A_m$, a joint reward function $\mathcal{R}$, a cost $\mathcal{C} = \{C_m\}, m = 1, \ldots, \mathcal{M}$ with safety bound d, a state transition function $\mathcal{T}$ and a discount factor γ. The goal of agents is to find a joint policy that maximizes the expected joint reward subject to a global constraint for expected cost function:

$$\begin{aligned}\max_{\pi\in\Pi_C}\ & J(\pi) = \mathbb{E}_{\tau\sim\pi}\left[\sum_t \gamma^t R(s_t, \mathbf{a}_t)\right]\\ s.t.\ & J_{C_m}(\pi) = \mathbb{E}_{\tau\sim\pi}\left[\sum_t \gamma^t C_m(s_t, \mathbf{a}_t)\right] \le d,\ \ m \in \mathcal{M}\end{aligned} \tag{11}$$

The detailed components are specified as below.

1) State and Observation: The environmental state $s_t = \{o_{1,t}, \ldots, o_{\mathcal{M},t}\} \in \mathcal{S}$ consists of the observations of all agents. The observation agent m can perceive is defined as

$$o_{m,t} = \left[p_{m,t}^{L}, p_{m,t}^{\text{RES}}, E_{m,t}^{\text{es}}, \lambda_t^{\text{FiT}}, \lambda_t^{\text{ToU}}, k_t^{b}, k_t^{s}, \mathbf{V}\right] \tag{12}$$

which consists of three parts: 1) Local information of load $p_{m,t}^{L}$, RES output $p_{m,t}^{\text{RES}}$ and SoC of ES $E_{m,t}^{\text{es}}$. 2) Market information of network tariff λ_t^{FiT}, λ_t^{ToU} and public orderbooks k_t^{b},k_t^{s}. Align with the real-world trading condition in [33], the agent has full access to the orderbook information, aimed at improving market responsiveness. 3) Grid information of voltage magnitude matrix $\mathbf{V}$. Note that $\mathbf{V}$ includes the voltage magnitude of the entire network, which is accessible to the agents through the measurement and communication devices such as phasor measurement units and wireless sensor networks. The similar setting can be found in the real-world MG [34] and research work [35].

2) Action: The action of agent m can be defined as

$$a_{m,t} = \left[a_{m,t}^{C}, a_{m,t}^{\text{MT}}, a_{m,t}^{\text{ES}}, a_{m,t}^{L}\right] \in \mathcal{A} \tag{13}$$

which comprises: $a^{C}_{m,t} \in [0,1]$ denoting the bidding price submitted to the auctioneer, normalized as a ratio of the difference between ToU and FiT $c_{m,t} = \lambda^{\text{FiT}}_t + a^{C}_{m,t}(\lambda^{\text{ToU}}_t - \lambda^{\text{FiT}}_t)$; $a^{\text{MT}}_{m,t}$, $a^{\text{ES}}_{m,t}$ and $a^{L}_{m,t}$ are actions controlling MT, ES and flexible load, respectively.

3) State Transition: The SOC transition of ES is determined by the action $a^{\text{ES}}_{m,t} \in [-1,1]$, where the positive value indicates charging and the negative value indicates discharging. The manually exclusive variables $p^{ch}_{m,t}$ and $p^{\text{dis}}_{m,t}$ are constrained by the range of energy capacity $[{}^{\text{es}}_{m}, \overline{E}^{\text{es}}_{m}]$, as shown below [12].

$$p^{ch}_{m,t} = \min\left(a^{\text{ES}}_{m,t}\overline{p}^{\text{es}}_{m}, \left(\overline{E}^{\text{es}}_{m} - E^{\text{es}}_{m,t}\right)/\eta^{ch}_{m}\Delta t\right) \quad (14)$$

$$p^{\text{dis}}_{m,t} = \max\left(a^{\text{ES}}_{m,t}\overline{p}^{\text{es}}_{m}, \left({}^{\text{es}}_{m} - E^{\text{es}}_{m,t}\right)\eta^{\text{dis}}_{m}/\Delta t\right) \quad (15)$$

Then, the SoC transition can be expressed as (6i).

4) Reward: The joint reward is defined as the summation of the negative energy cost of MGs in equation (6a):

$$R_t = -\sum_{m=1}^{\mathcal{M}} H_{m,t} \quad (16)$$

5) Cost: The DSO runs the power flow to derive the node voltage profile. The cost function is designed as the total penalization for voltage violations. Since the line capacity is sufficient in DN, the line congestion is not considered. Similar setting can be found in [36].

$$C_{m,t} = \sum_{i=1}^{W}\left(\left[V_{i,t} - V^{+}_{i}\right]^{+} + \left[V^{-}_{i} - V_{i,t}\right]^{-}\right) \quad (17)$$

where $[\,\cdot\,]^{+/-}$ denotes $max/min\{\cdot, 0\}$,the square brackets represent the penalty level for voltage violation. Each DSO-owned agent has a local cost network to estimate the global violation penalty C, so $C_{m,t} = C,\ \forall m \in \mathcal{M}$.

III. Proposed Decentralized Multi-Agent Safety-Aware Learning Method

In this section, we propose a decentralized global constraint multi-agent safety-aware learning (MASAL) method to solve the Constrained Cooperative Markov Game. To remedy the poorly coordinated trading policy when agents update at the same time, the proposed method conducts policy update sequentially. To tackle the agents' unawareness of the safety level of policy update, an additional cost value network is introduced. This guarantees the performance in optimal trading policy search under the safety concern.

A. Challenges of Multi-Agent Safe Policy Optimization

Considering a P2P trading problem with decentralized agents, it is challenging to find the coordinated policy owing to conflicting interaction of agents. 1) Given the profit and safety violation cost determined by joint policy, an individual agent may struggle to distill its own contribution. 2) Even when an agent successfully identifies its policy update direction, it may conflict with the other agents due to miscoordination. 3) Conditions where the goals of profit improvement and constraint satisfaction conflict can occur. In such conditions, policy optimization should prioritize safety. Next, we will illustrate these challenges in Fig. 1 by taking a two-player P2P trading differential game with total energy profit $f_1(a^1, a^2)$ and constraint function $f_2(a^1, a^2) < d$ as an example. Contours of the total energy profit and constraint value are plotted. The area within the constraint contour at value d is the safe region, while policies outside this contour are unsafe. The x and y axes represent the agent's actions, with positive values for selling energy and negative values for buying energy.

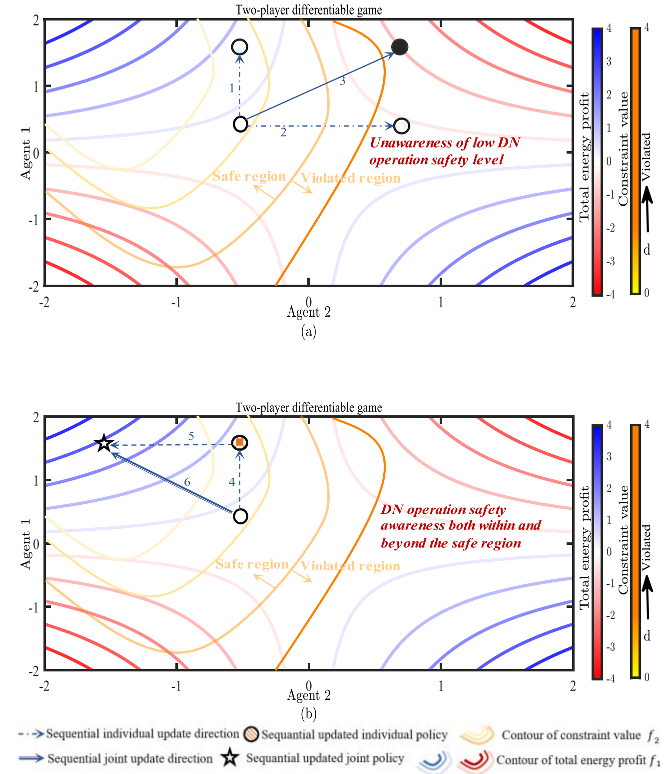

Fig. 1. (a) Illustration of agents' potential conflicts regarding energy profit improvement and safety constraint satisfaction faced by current safe RL methods. (b) Illustration of desirable safety-aware joint policy optimization method.

It is assumed that the initial joint trading policy is in the second quadrant with a constraint value closed to the upper limit d. When conducting simultaneous trading policy update to increase the profit as in Fig. (1a), agent 1 produces more surplus power and wishes to sell it (direction 1). Meanwhile, for the same purpose, agent 2 curtails its energy consumption to become a seller as well (direction 2). However, extra energy is generated but energy deficit is vanished, so their trading policy updates are conflicting, leading to a decreased total energy profit. More importantly, as there is no violation penalty at the beginning, two agents remain unaware of initial joint trading policy's large potential for DN constraint violation during update. This unawareness endangers the safety of DN operation (direction 3). In contrast, the desirable update scheme should be safety-aware both within and beyond the safe region as in Fig. (1b), so safety cost gradient value can be considered together with energy profit gradient. Safety awareness can give agents foresight, preventing them from blindly searching for profit improvement policy until violation happens. Also, sequentially updating individual policy would avoid the potential conflict among agents. For example, agent 1 generates extra energy as a seller (direction 4) first. And agent 2 imports more energy from agent 1 as a buyer (direction 5) afterward. The derived joint policy realizes profit improvement and constraint satisfaction concurrently (direction 6).

B. Multi-Agent Advantage Decomposition

Different from [12], [37], we further evaluate the action's influence on system operation. Sequential update [38] is a promising way. It enables agents to consider the impact of the previous agents' bidding decisions on the system safety violation and trading profits. Therefore, the agents' policy update can be efficiently coordinated. Let $\theta^m, m \in \mathcal{M}$ parameterize each agent's policy neural network, thus the joint policy π is parameterized by $\theta = (\theta^1, \ldots, \theta^{\mathcal{M}})$.

We first define state-action value function for cost as $Q_\pi^C(s, \mathbf{a}) = \mathbb{E}_{\tau\sim\pi}[R(\tau_\pi)|s_0 = s, \mathbf{a}_0 = \mathbf{a}]$, denoting the expected cost from state s, taking action $\mathbf{a}$ under policy π. The state value function for cost is $V_\pi^C(s) = \mathbb{E}_{\tau\sim\pi}[Q_\pi^C(s, \mathbf{a})]$, denoting the expected cost from state s under policy π. On the other hand, the state-action and state value function for reward can be written as Q_π and V_π, respectively.

To evaluate unique contribution from a certain part of the agents $m_{1:h}$ while marginalizing out the rest $n_{1:k}$, we define the multi-agent advantage function as

$$A_\pi^{m_{1:h}}\left(s, \mathbf{a}^{n_{1:k}}, \mathbf{a}^{m_{1:h}}\right) \triangleq Q_\pi^{n_{1:k}, m_{1:h}}\left(s, \mathbf{a}^{n_{1:k}}, \mathbf{a}^{m_{1:h}}\right) - Q_\pi^{n_{1:k}}\left(s, \mathbf{a}^{n_{1:k}}\right) \quad (18)$$

where $m_{1:h}$ represents an ordered subset of $\{m_1, \ldots, m_h\}$ of agents $\mathcal{M}$ and $n_{1:k}$ refers a disjointed subset $\{n_1, \ldots, n_k\}$. $Q_\pi^{n_{1:k}, m_{1:h}}(s, \mathbf{a}^{n_{1:k}}, \mathbf{a}^{m_{1:h}})$ denotes the state-action value function of agents $n_{1:k}, m_{1:h}$ taking action $\mathbf{a}^{n_{1:k}}, \mathbf{a}^{m_{1:h}}$, and $Q_\pi^{n_{1:k}}(s, \mathbf{a}^{n_{1:k}})$ denotes the state-action value function of agents $n_{1:k}$ taking action $\mathbf{a}^{n_{1:k}}$. $A_\pi^{m_{1:h}}(s, \mathbf{a}^{n_{1:k}}, \mathbf{a}^{m_{1:h}})$ evaluates the advantage of agents $m_{1:h}$ taking actions $\mathbf{a}^{m_{1:h}}$ under state s provided that agents $n_{1:k}$ taking actions $\mathbf{a}^{n_{1:k}}$. The definition holds for both cost advantage function A_π^C and reward advantage function A_π. The multi-agent advantage functions build the theoretical foundation for decomposing the joint trading profit and safety cost as a sum of sequentially-unfolding profit and safety cost of individual agents as below.

Proposition 1 (Multi-Agent Advantage Decomposition). Let π be a joint policy and $m_{1:h}$ be a random ordered agent subset. For any state s and joint action $\mathbf{a}^{m_{1:h}}$

$$A_\pi^{m_{1:h}}\left(s, \mathbf{a}^{m_{1:h}}\right) = \sum_{j=1}^{h} A_\pi^{m_j}\left(s, \mathbf{a}^{m_{1:j-1}}, a^{m_j}\right) \quad (19)$$

Proposition 1 verifies that the sequential update is a valid method for seeking trading profit improvement joint actions with safety concern in a decentralized way. Specifically, consider agents taking actions sequentially in an arbitrary order $m_{1:n}$ to avoid the potential conflict. Considering the previous agents' actions, each agent chooses an action a^{m_h} such that $A_\pi^{m_h}(s, \mathbf{a}^{m_{1:h-1}}, a^{m_h}) > 0$ and $A_\pi^{C,m_h}(s, \mathbf{a}^{m_{1:h-1}}, a^{m_h}) < 0$. According to proposition 1, the derived joint action ensures a positive $A_\pi(s, \mathbf{a})$ and a negative $A_\pi^C(s, \mathbf{a})$, thereby ensuring profit improvement and safety violation reduction at each iteration. The miscoordination of agents is remedied.

Proof: **Step 1:** Definition of the Advantage Function

$$A_\pi^{m_{1:h}}\left(s, \mathbf{a}^{m_{1:h}}\right) = Q_\pi^{m_{1:h}}\left(s, \mathbf{a}^{m_{1:h}}\right) - V_\pi(s) \quad (20)$$

where $Q_\pi^{m_{1:h}}(s, \mathbf{a}^{m_{1:h}})$ is the joint action value function for the subset $m_{1:h}$ and $V_\pi(s)$ is the state value under the policy π.

Step 2: Recursive Expansion of the Q-Function

The Q-function satisfies recursive decomposition:

$$Q_\pi^{m_{1:h}}\left(s, \mathbf{a}^{m_{1:h}}\right) = \sum_{j=1}^{h} Q_\pi^{m_{1:j}}\left(s, \mathbf{a}^{m_{1:j}}\right) - Q_\pi^{m_{1:j}}\left(s, \mathbf{a}^{m_{1:j-1}}\right) \quad (21)$$

This step is justified by the property that the Q-function of a subset of agents can be recursively expanded as a sum of marginal contributions from individual agents.

Step 3: Reformulating in Terms of Advantage Functions

Using the definition of the advantage function (20), each term is rewritten in the sum as

$$Q_\pi^{m_{1:j}}\left(s, \mathbf{a}^{m_{1:j}}\right) - Q_\pi^{m_{1:j-1}}\left(s, \mathbf{a}^{m_{1:j-1}}\right) = A_\pi^{m_j}\left(s, \mathbf{a}^{m_{1:h-1}}, a^{m_j}\right) \quad (22)$$

Equation (22) aligns with the equation (18) in the manuscript. Thus, substituting this into the summation as:

$$A_\pi^{m_{1:h}}\left(s, \mathbf{a}^{m_{1:h}}\right) = \sum_{j=1}^{h} A_\pi^{m_j}\left(s, \mathbf{a}^{m_{1:j-1}}, a^{m_j}\right) \quad (23)$$

which completes the proof. ■

C. Multi-Agent Safety-Aware Learning Algorithm

To stabilize trading policy with safety concern, a trust region constraint is introduced to ensure the KL divergence between the new policy π_{k+1} and old policy π_k does not exceed a preset threshold δ, thereby balancing exploration and exploitation [39]. Thus, the proposed algorithm solves the Constrained Markov Game by maximizing the reward within a feasible policy set Π_θ.

$$\begin{aligned} \pi_{k+1} &= \arg\max_{\pi\in\Pi_\theta} \; J(\pi) \\ s.t. \quad & J_C(\pi) \le d \\ & D_{\mathrm{KL}}^{\max}(\pi \,||\, \pi_k) \le \delta \end{aligned} \quad (24)$$

It is impractical to search for $D_{\mathrm{KL}}^{\max}(\pi \,||\, \pi_k)$ as it necessitates evaluating the KL-divergence for all states at each iteration. With stochastic sampling method, the expected KL-divergence $\overline{D}_{\mathrm{KL}}(\pi \,||\, \pi_k)$ is used to substitute the maximal KL-divergence. To closely approximate (24), surrogate functions from CPO [40] are used to substituted the objective and safety constraints.

$$\begin{aligned} \pi_{k+1} &= \arg\max_{\pi\in\Pi_\theta} \; \mathbb{E}_{\substack{s\sim d^{\pi_k} \\ a\sim\pi}}\left[A_{\pi_k}(s, \mathbf{a})\right] \\ s.t. \quad & J_C(\pi_k) + \mathbb{E}_{\substack{s\sim d^{\pi_k} \\ a\sim\pi}}\left[A_{\pi_k}^C(s, \mathbf{a})\right] \le d \\ & \overline{D}_{\mathrm{KL}}(\pi \,||\, \pi_k) \le \delta \end{aligned} \quad (25)$$

To further reduce the computation burden in (25) caused by the policy with high-dimensional parameter spaces, we can approximate (25) by using a first-order Taylor expansion for the objective and global cost constraint, and a second-order expansion for the KL divergence constraint as

$$\begin{aligned} \theta_{k+1} &= \arg\max_{\theta} \ g^T(\theta - \theta_k) \\ s.t. &\quad c + b^T(\theta - \theta_k) \leq 0 \\ &\frac{1}{2}(\theta - \theta_k)^T \mathbf{H}(\theta - \theta_k) \leq \delta \end{aligned} \tag{26}$$

where $g \triangleq \nabla_\theta \mathbb{E}[A_{\pi_k}(s, \mathbf{a})]$ denotes the gradient of the reward objective, $c \triangleq J_C(\pi_k) - d$ denotes the difference between the safety cost's expectation $J_C(\pi_k)$ and the safety bound d, $b \triangleq \nabla_\theta \mathbb{E}[A^C_{\pi_k}(s, \mathbf{a})]$ denotes the gradient of global safety cost, and $\mathbf{H}$ is the Hessian of KL divergence.

The above policy optimization problem (26) is convex since $\mathbf{H}$ is always positive semi-definite. It can be solved through the dual form as written in (27).

$$\max_{\lambda \geq 0, v \geq 0} -\frac{1}{2\lambda}\left(g^T \mathbf{H}^{-1} g - 2r^T v + v^T \mathrm{S} v\right) + v^T c - \frac{\lambda\delta}{2} \tag{27}$$

where $r \triangleq g^T \mathbf{H}^{-1}$b and S $\triangleq b^T \mathbf{H}^{-1} b$, solely to simplify the notation.

Given that λ^*, v^* are the solution to the dual form, the policy update direction is determined through the conjugate gradient method, as in Eq. (28).

$$x_k = \mathbf{H}^{-1}(g - bv^*) \tag{28}$$

Then, the current policy can be updated according to (29).

$$\theta_{k+1} = \theta_k + \frac{\alpha}{\lambda^*} x_k \tag{29}$$

To determine the step size α of the above update, backtracking line search starting at $1/\lambda^*$ is used.

When the convex optimization problem is infeasible, appropriate neural network parameters cannot be found to keep the safety cost function below the preset safety bound and improve the reward at the same time. Consequently, we seek a semi-optimal policy that minimizes the safety cost. To achieve the global minimum of the safety cost function, we disregard the reward improvement direction's effects during policy updates:

$$\begin{aligned} \theta_{k+1} &= \arg\min_{\theta} \left[c + b^T(\theta - \theta_k)\right] \\ s.t. &\ \frac{1}{2}(\theta - \theta_k)^T \mathbf{H}(\theta - \theta_k) \leq \delta \end{aligned} \tag{30}$$

The dual form of (30) is $L(x, \lambda) = c + b^T x + \lambda(\frac{1}{2}x^T \mathbf{H} x - \delta)$. Let $x = \theta - \theta_k$ denotes the policy change. When the partial derivative of the dual function concerning the policy change is 0, the optimal update is achieved.

$$\frac{L}{\partial x} = b^T + \lambda(\mathbf{H}x) = 0 \tag{31}$$

The obtained solution $x = -\frac{1}{\lambda}\mathbf{H}^{-1} b^T$ should satisfy the trust region constraint as below.

$$\frac{1}{2} x^T \mathbf{H} x \leq \delta \tag{32}$$

As a result, the inequality $\sqrt{\frac{b^T \mathbf{H}^{-1} b}{2\delta}} \leq \lambda$ holds and the update rule under infeasible condition is:

$$\theta_{k+1} = \theta_k - \sqrt{\frac{2\delta}{b^T \mathbf{H}^{-1} b}} \mathbf{H}^{-1} b \tag{33}$$

Eqs. (29) and (33) are employed to implement the MASAL.

D. Extension to General System Operation Conditions

There is the other drawback we want to highlight in conventional safe RL methods. They cannot distinguish whether the safety violation is caused by the action. When the violation originates from exogenous environments rather than endogenous actions, they cannot effectively conduct a responsive policy.

To extend the above method to general system operation conditions and realize safety-aware, the safety level of current policy as well as next policy should be thoroughly considered. Three important metrics below are defined to comprehensively evaluate the safety level, serving as the foundation for system-wide safety condition classification.

We define the safety cost function's policy gradient estimation as the **safety cost gradient** in Eq. (34).

$$\widehat{b} = \nabla_\theta J_C(\pi_k) = \nabla_\theta \mathbb{E}\left[\sum_t \gamma^t C\left(s^t, \mathbf{a}^t, s^{t+1}\right)\right] \tag{34}$$

The **safety margin**, estimating the distance between the safety cost and safety bound under the current policy, is defined in Eq. (35). A positive and closed-to-zero value of $\widehat{c}$ means a sufficient safety margin of current policy, while a negative value of $\widehat{c}$ means an already dangerous current policy.

$$\widehat{c} = \left(d - \mathbb{E}\left[\sum_t \gamma^t C\left(s^t, \mathbf{a}^t, s^{t+1}\right)\right]\right) \cdot (1 - \gamma) \tag{35}$$

The **safe recovery feasibility** is defined in Eq. (36) to judge whether it is feasible to return to safe region from a dangerous situation through policy update. A positive F means the KL divergence-constrained trust region does not intersect with the safety bound. The non-intersection arises from two cases: either the entire trust region is safe or the entire trust region is dangerous. A negative F means that the KL divergence-constrained trust region intersects with the safety bound, thus dividing the trust region into safe and unsafe parts.

$$F = \frac{\widehat{c}^2}{\widehat{b}^T \mathbf{H}^{-1} \widehat{b}} - 2\delta \tag{36}$$

According to the above definitions, the conditions are introduced in order from the safest to the most dangerous.

Normal Condition 1: If the safety margin $\widehat{c} > 0$ and the safety cost gradient $\widehat{b} \leq 10^{-8}$, it indicates a currently safe policy and a non-increasing safety cost. Even in the absence of safety cost constraint, policy update solely based on KL divergence constraint will not pose a safety risk. The policy update under condition 1 is at an **extremely high safety level**.

Normal Condition 2: If $\widehat{c} > 0$ and the safe recovery feasibility $F > 0$, it indicates a currently safe policy and that the whole trust region is within the safety bound. The policy update under condition 2 is at a **high safety level**.

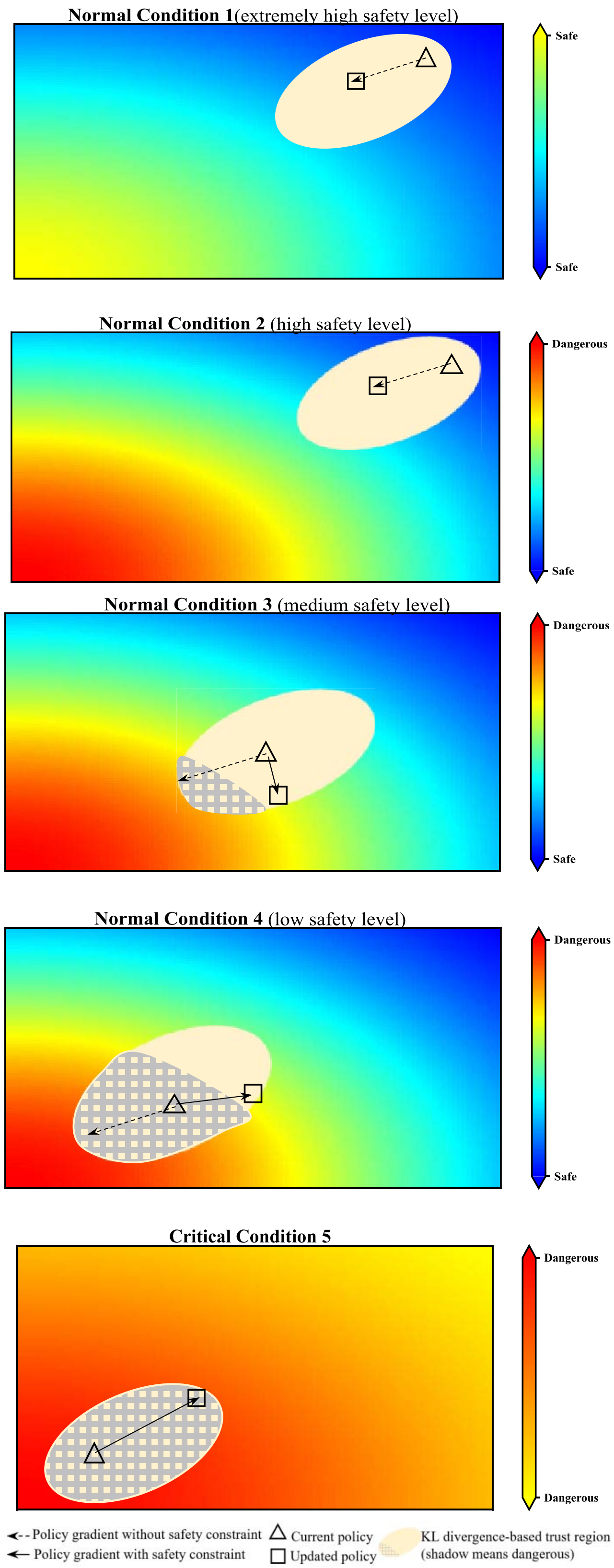

Fig. 2. System-wide safety condition classification.

When detecting system condition 1 or 2, the DN has sufficient flexibility resources to support the P2P transaction. Under conditions when only the gradient of energy profits matters, the policy update can straightforwardly employ the conventional Trust Region Policy Optimization (TRPO) technique [39]:

$$\theta_{k+1} = \theta_k - \sqrt{\frac{2\delta}{\widehat{g}^T \mathbf{H}^{-1} \widehat{g}}} \mathbf{H}^{-1} \widehat{g} \tag{37}$$

Normal Condition 3: If the safety margin $\widehat{c} > 0$ and the safe recovery feasibility $F \leq 0$, it indicates the current policy is safe and the trust region is divided into safe and dangerous parts by the safety bound. The policy update under condition 3 is at a **medium safety level**.

Normal Condition 4: If $\widehat{c} \leq 0$ and $F \leq 0$, it indicates a currently dangerous policy and the updated policy can return to safe region due to the intersection. The policy update under condition 4 is at a **low safety level**.

When detecting system condition 3 or 4, improving trading profits may lead to violation, so P2P transaction should be restricted for compliance of network constraint. It is feasible to recover a currently dangerous policy to a safe one through policy update due to the trading capacity. The policy updates towards the joint direction of trading profit improvement and safety cost reduction, as in Eq. (29).

Critical Condition 5: If the safety margin $\widehat{c} \leq 0$ and the safe recovery feasibility $F > 0$, it indicates a currently dangerous policy and that the whole trust region is out of the safety bound. It is **infeasible** to find the policy region satisfying both KL divergence and safety constraints, so the updated policy cannot return to the safe region. Under this critical condition, we conduct **responsive** trading policy and seek safety cost minimization.

When detecting system condition 5, the violation stems from exogenous DN operation rather than endogenous transaction, which cannot be completely eliminated through trading policy update. To achieve safety-aware policy search, the policy updates towards the steepest decrease in safety cost, as illustrated in Eq. (33). The reward improvement is not considered under this condition.

IV. NUMERICAL RESULTS

The proposed Multi-Agent Safety-Aware Learning algorithm is validated via the IEEE 33-bus distribution system as depicted in Fig. 3. The MGs are positioned at bus 10, 18, 22 and 25. RES output and load profile of MMGs are sourced from [41]. The dataset spans a period of ten days. To replicate real-world variability, randomness is incorporated into the energy demand and renewable generation during the training process. And a 20% prediction error is considered. The allowable range of voltage is [0.95,1.05]. Sourced from [42], the FiT and ToU rates are structured based on different time periods throughout the day, as shown in Table II. The GPU is NVIDIA GeForce RTX 4060, and the operating system is Ubuntu 20.04.

To validate the efficiency of proposed MASAL algorithm, its performance is benchmarked against four baseline methods, including

1) *Multi Agent Proximal Policy Optimization (MAPPO) [43]:* Voltage magnitude violation is not considered during in the trading.

Algorithm 1 Multi-Agent Safety-Aware Learning (MASAL)

Initialize: Actor networks $\{\theta_0^m, \forall m \in \mathcal{M}\}$, Global V-value network $\{\phi_0\}$, Individual V-cost networks $\{\phi_{C_m0}, \forall m \in \mathcal{M}\}$, Replay buffer $\mathcal{D}$
For $k = 0$ to $K-1$ **do**
  Collect trajectories $\tau_{\pi_k} = (s_t, \mathbf{a}_t, r_t, c_t, s_{t+1})$ by running the joint policy
  Calculate reward and cost advantage functions: A_π, A_π^C
  Calculate reward gradient $\widehat{g}$, safety cost gradient $\widehat{b}$, safety margin $\widehat{c}$ and safe recovery feasibility F
  Generate the random sequence for MMGs' bidding
  For agent $m = 1$ to $\mathcal{M}$ **do**
    If condition 1 or condition 2 **then**
      Update policy: $\theta_{k+1} = \theta_k - \sqrt{\frac{2\delta}{\widehat{g}^T \mathbf{H}^{-1}\widehat{g}}}\mathbf{H}^{-1}\widehat{g}$
    Elseif condition 3 or condition 4 **then**
      Get solution λ^*, v^* to the dual problem of Eq. (27)
      Solve step size α by backtracking line search
      Update policy: $\theta_{k+1} = \theta_k + \frac{\alpha}{\lambda^*}\mathbf{H}^{-1}(\widehat{g} - \widehat{b}v^*)$
    else
      Update policy: $\theta_{k+1} = \theta_k - \sqrt{\frac{2\delta}{\widehat{b}^T \mathbf{H}^{-1}\widehat{b}}}\mathbf{H}^{-1}\widehat{b}$
    End if
  End for
  Update individual cost value networks:

$$\phi_{C_m} = \arg\min_{\phi} \mathbb{E}\left[\left(V_{\phi_{C_m}}(s_t) - \widehat{C}_{m,t}\right)^2\right]$$

  Update global reward value network:

$$\phi_R = \arg\min_{\phi} \mathbb{E}\left[\left(V_{\phi_R}(s_t) - \widehat{R}_t\right)^2\right]$$

End for

2) *MAPPO-PF [18]:* A violation penalty function with fixed penalty factor is added to the reward.

3) *MAPPO-SL [44]:* A QP-based safety layer is used to project dangerous actions to the closest (Euclidean distance) safe ones. It contains two processes: The "Safety examination" includes a pretrained NN serving as a regressor predicting the state transition of voltage magnitudes. The "Safety projection" solves an QP problem with CVXPY.

$$\underset{\mathbf{a}_t'}{\text{argmin}}\ \|\mathbf{a}_t' - \mathbf{a}_t\|^2 \quad (38)$$

$$s.t.\ f(s_{t-1}, \mathbf{a}_t) \leq \Delta V \quad (39)$$

Let $\mathbf{a}_t$ denote the original joint action, and $\mathbf{a}_t' = \mathbf{a}_t + \Delta\mathbf{a}_t$ denotes the joint action modified by the safety layer. The objective (38) aims to find a perturbed joint action $\mathbf{a}_t'$ that differs the least from the original joint action $\mathbf{a}_t$. $f(s_{t-1}, \mathbf{a}_t)$ is the approximated value of maximum voltage deviation $\max_i |V_{i,t}(s_{t-1}, \mathbf{a}_t) - 1|$ and ΔV is set as 0.05 p.u. in this study.

4) *MAPPO-Lag [23]:* A Lagrangian multiplier-based penalty is integrated in the joint reward.

5) *Constrained Policy Optimization (CPO) [27]:* This baseline method assumes that all MGs are managed by a single operator, and a single-agent safe DRL algorithm is employed to address the voltage constraint.

TABLE II
GRID PRICE STRUCTURE WITH TOU AND FIT RATES

Time Period	ToU ($/kWh)	FiT ($/kWh)
9:00-16:00 (Shoulder)	0.13	0.03
17:00-20:00 (Peak)	0.18	0.04
21:00–8:00 next day (Off-peak)	0.08	0.02

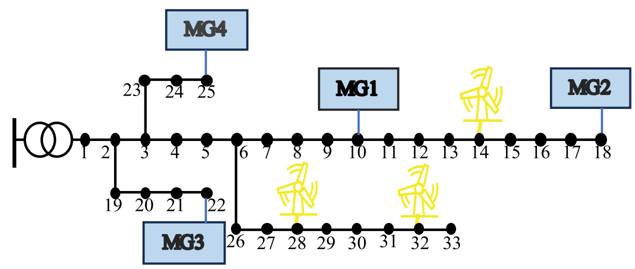

Fig. 3. Physical topology for MMGs' P2P transaction.

TABLE III
HYPER PARAMETERS REGARDING TO ALGORITHM IMPLEMENTATIONS

Algorithm	Hyper Parameters
Proposed MASAL	KL threshold: 0.09 Safety bound: 0.1 Line search fraction: 0.5 Fraction coefficient: 0.1 Learning rate: 5e-4 Discount factor: 0.99
MAPPO	Optimizer: Adam Episode length: 24 Hidden sizes: [128,128] Batch size: 256 Clip rate: 0.2 PPO epoch: 5
MAPPO-PF	Penalty factor: 50
MAPPO-SL	Similar to [44], approximate the voltage violation cost as the product of non-linear features learned from the state and actions.
MAPPO-Lag	Lagrangian coefficient: 0.4 Lagrangian coefficient learning rate: 0.0005
CPO	In order to keep the fairness, all the hyper parameters are same as those in the MASAL algorithm.

6) *Conventional Zero Intelligence (ZI) Strategy [45]:* A fundamental trading strategy in the RDA market. ZI selects the price bid uniformly at random values between FiT and ToU. Hyperparameters regarding to algorithm implementations are listed in Table III. The performance indicators of the comparison contain two aspects: 1) The economic indicator, i.e., energy cost; 2) The operational/safety-related indicator, i.e., the safety cost (constraint violation cost).

A. Performance Comparison Under Normal Conditions 1-4

To compare the proposed MASAL and baseline methods under normal conditions, the average performance of each method is assessed periodically (every 30 episodes) during training as shown in Fig. 4. It can be observed that the proposed algorithm achieves 0 safety violation during training

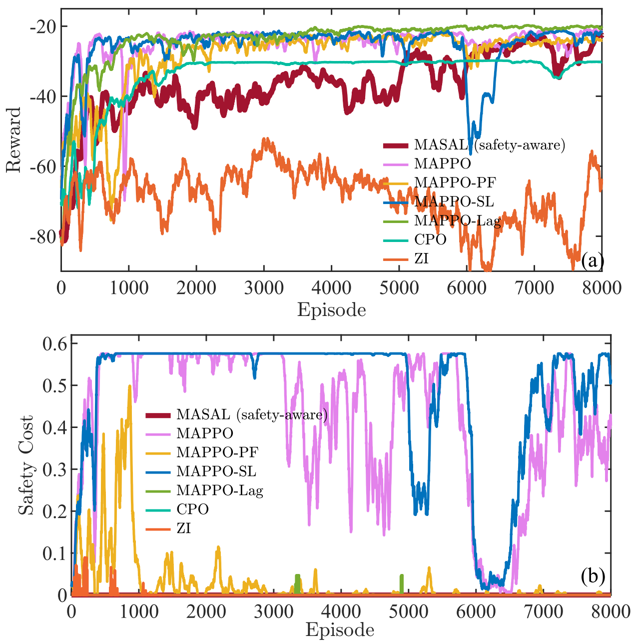

Fig. 4. Training performance comparison under normal conditions in 33-bus distribution system (a) Episode reward (b) Episode cost.

and outperforms the other methods. And the obtained reward is 98.64% of that obtained by MAPPO without safety constraint. It results from integrating safety cost gradient into policy update direction and conducting preventive trading policies under the proposed MASAL method. MAPPO-Lag still leads to violations from 1st to 6000th episode and eliminates voltage violation after 6000-episode training. MAPPO-PF cannot guarantee safety because the fixed penalty factor cannot adapt to the constantly changing value of safety cost and reward. Violation cost under MAPPO-SL method remains even 19% higher than MAPPO at the end of training process. The reasons can be attributed to two main factors: First, the safety layer method assumes a linear relationship between the state transition of voltage magnitude and the action, which is not applicable in the network-constrained P2P trading problem. The prediction error in the state transition of voltage magnitude renders the examination step of the safety layer method ineffective. Second, when the joint action is identified as dangerous, the action modification is performed at a single time step, which can lead to time-coupling constraint violations.

Well-trained agents are employed to generate trading decisions over an operational day. Regarding the economic efficiency of learned MMG bidding policies during testing, as shown in Table IV, energy costs of MMGs under MASAL are 101.37% of that obtained by MAPPO without safety constraint, and the P2P traded quantity is the highest. The reason is that the MMG might reschedule their supply and demand for compliance of voltage constraints.

Taking 12:00, when the voltage violation occurs, as an example, the voltage magnitude in Fig. 5 and the trading outcomes in Table V are analyzed. The voltage at nodes 11, 12 and 13 exceed the upper limit under MAPPO and MAPPO-SL.

TABLE IV
AVERAGE TRADING QUANTITIES AND ENERGY COSTS OF MMGS IN AN OPERATIONAL DAY DURING TESTING

Algorithm	P2P (kWh)	Energy Costs ($)
MASAL	**6043**	**221**
MAPPO	5702	218
MAPPO-PF	5701	229
MAPPO-SL	5472	222
MAPPO-Lag	5209	207
CPO	4653	301
ZI	2682	465

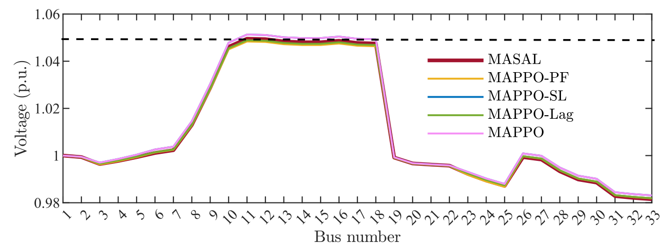

Fig. 5. Voltage magnitude at 12:00 under normal condition.

TABLE V
TRADING OUTCOMES AT 12:00 UNDER NORMAL CONDITIONS

Algorithm	Seller	Buyer	P2P Quantity (kWh)
MASAL	MG1	MG3	905
	MG1	MG4	453
MAPPO-SL	MG1	MG3	830
	MG1	MG2	428

Comparing the trading policies under MASAL with MAPPO-SL, some surplus power of MG1 is traded with MG4 instead of MG2. This is because the power flow from MG1 (node 10) to MG2 (node 18) would cause the voltage, which is already close to the upper limit, to exceed it. Meanwhile, selling MG1's power to MG4 does not exacerbate this danger. This indicates that MASAL can effectively capture the highly non-linear and indirect relationship between trading policy and safety violation cost. The safety-aware feature also enables MMG to adjust trading policies on the verge of violating voltage constraints back to a relatively safe region.

B. Performance Comparison Under Critical Conditions 5

To assess the performance of the proposed method under critical conditions, we assume that the power generation of the WT at node 14 has increased due to weather conditions, leading to exogenous voltage violations. It cannot be completely eliminated by adjusting the joint trading policy owing to limited trading capacity. Safety bound of MASAL is reset as 1 with all the other parameters remaining unchanged. Fig. 6 depicts the training performance under critical conditions. According to system safety classification in Section III, condition 5 occurs a total of 196 times. During 8000 training episodes the safety cost decreases almost monotonically and

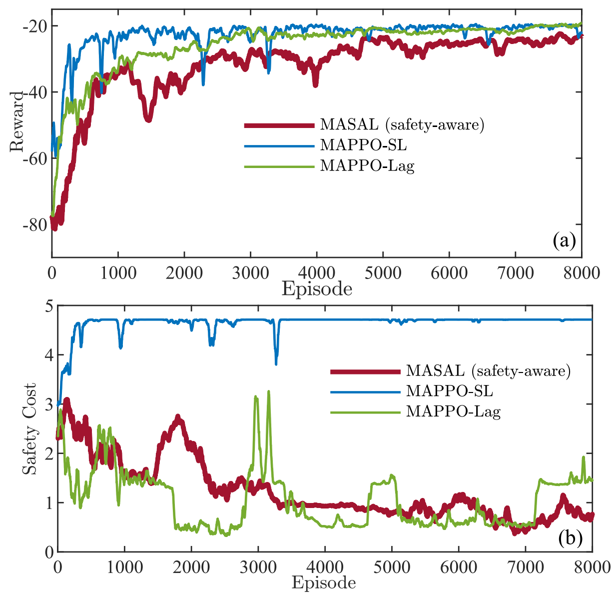

Fig. 6. Training performance comparison under critical conditions in 33-bus distribution system (a) Episode reward (b) Episode cost.

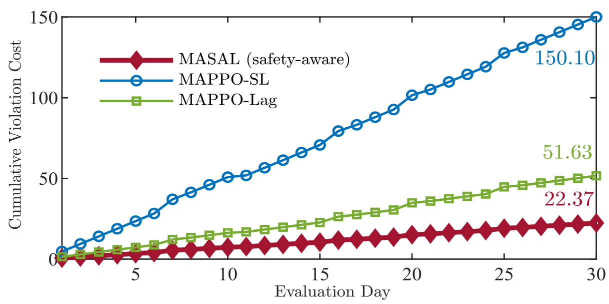

Fig. 7. Cumulative violation cost under critical conditions.

gradually falls below the safety bound after 6000 episodes under the proposed method. This is because the exogenous violations from DN operation are effectively detected and then responsive trading policies are conducted. Due to the safety-aware nature, the reward occasionally decreases but ultimately reaches 86.4% of that obtained by the MAPPO-SL. The safety layer method does not work because the QP-based projection problem has no feasible solution. Unlike normal conditions, MAPPO-Lag cannot steadily decrease the safety cost under critical conditions. The main reason is that integrates safety performance and profit performance metrics into the reward function complicates the learning process.

To evaluate the test performance of the three safe RL methods, their trained online actor networks are loaded to derive the trading actions. Fig. 7 presents the cumulative violation cost under critical conditions over the 30 evaluation days. It can be observed that cumulative violation cost under MASAL is the lowest. It is 56.15%, 85.10% lower than that under MAPPO-Lag, MAPPO-SL, respectively.

Voltage magnitude and trading outcomes are illustrated in Fig. 8 and Table VI, respectively. Trading policy under the proposed MASAL results in the fewest voltage violation nodes, followed by MAPPO-Lag, with MAPPO-SL causing

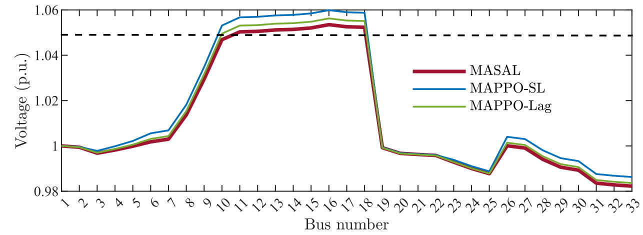

Fig. 8. Voltage magnitude at 12:00 under critical condition.

TABLE VI
TRADING OUTCOMES AT 12:00 UNDER CRITICAL CONDITIONS

Algorithm	Seller	Buyer	P2P Quantity (kWh)
MASAL	MG1	MG3	830
	MG1	MG4	427
MAPPO-Lag	MG1	MG3	878
	MG1	MG4	279
	MG2	MG4	109

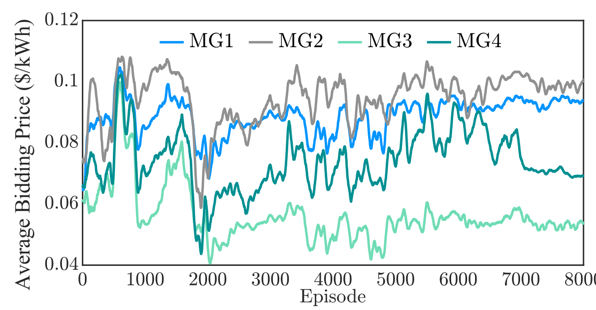

Fig. 9. Convergence performance of average bidding price under MASAL algorithm.

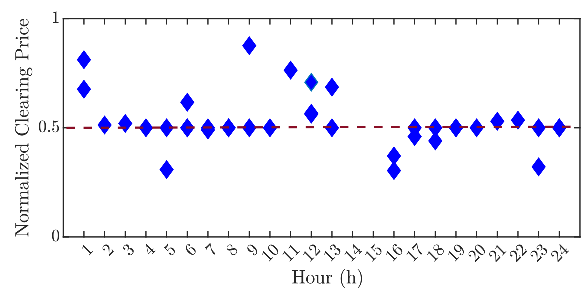

Fig. 10. Normalized clearing price under MASAL algorithm.

the most voltage violation nodes. Comparing the trading outcomes of MASAL with MAPPO-Lag, it can be seen that MG4 chooses to purchase the deficient power from MG1 instead of MG2. This is because the power flow from MG2 (node 18) to MG4 (node 25) will further increase the voltage magnitude of nodes 11-18. This indicates that even when reward improvement and safety cost reduction cannot be simultaneously achieved, the proposed method can identify the occurrence of critical conditions and prioritize the reduction of safety cost.

C. Bidding Action Analysis of MMG Agents

Fig. 9 illustrates the convergence performance of average bidding prices of agents per day under the MASAL algorithm.

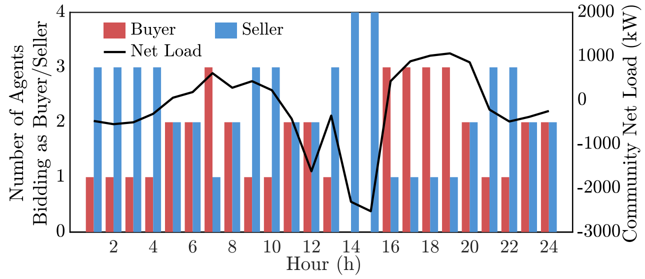

Fig. 11. Number of agents bidding as buyer/seller and community net load during the operational day.

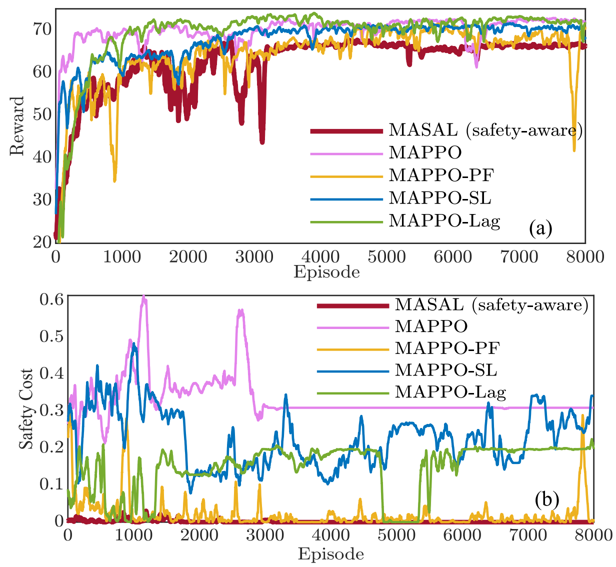

Fig. 12. Training performance comparison under normal conditions in 123-bus distribution system (a) Episode reward (b) Episode cost.

It shows that each agent adjusts its bidding strategy to adapt to the strategies of the other agents, and their joint bidding strategy stabilizes and converges around the 7000th episode. For a more in-depth analysis, Fig. 10 demonstrates the clearing price at the test stage. The P2P clearing price is transformed into normalized clearing price $\lambda_{i,t}^{m} - \lambda_{n,t}^{\text{FiT}}/(\lambda_{n,t}^{\text{ToU}} - \lambda_{n,t}^{\text{FiT}})$ for clarity, where a normalized clearing price closer to 1 indicates that the actual clearing price aligns more closely with ToU and a value closer to 0 indicates alignment with FiT. Fig. 11 shows the number of agents bidding as buyer/seller and community net load during the operational day. At 1:00 and 3:00, the P2P clearing price approaches the FiT, as the number of agents bidding as sellers exceeds the number of agents bidding as buyers, indicating a buyer's market. At 14:00 and 15:00, the renewable energy generation in the MMG community is the highest, and all agents are bidding as sellers, resulting in no successful peer-matching. Between 16:00 and 18:00, the number of agents bidding as buyers exceeds those bidding as sellers, creating a seller's market, which leads to higher transaction prices, approaching the ToU rate.

D. Scalability and Computational Performance Analysis

In the 123-bus system, eight PVs are assumed to be installed at nodes (4, 27, 42, 57, 64, 70, 87, 97) [46]. Eight MGs are located at nodes (50, 53, 62, 76, 80, 89, 99, 106). Fig. 12 depicts the training performance comparison under normal conditions in 123-bus distribution system. The PV output at node 57 is artificially increased to create critical conditions, and the corresponding training performance is shown in Fig. 13. It can be seen that, in the case of a larger distribution system and a greater number of agents, the proposed MASAL method still outperforms the other methods, demonstrating scalability.

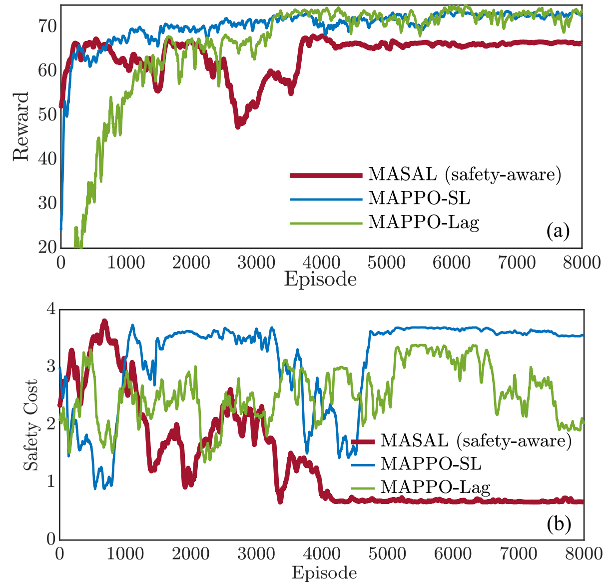

Fig. 13. Training performance comparison under critical conditions in 123-bus distribution system (a) Episode reward (b) Episode cost.

TABLE VII
COMPUTATIONAL COMPLEXITY COMPARISON

Algorithm	Computational complexity
MASAL (proposed)	$O(M^2 \cdot \sum_{m\in\mathcal{M}} \lvert\mathcal{S}_m\rvert \cdot \lvert\mathcal{A}_m\rvert)$
MAPPO	$O(\sum_{m\in\mathcal{M}} \lvert\mathcal{S}_m\rvert \cdot \lvert\mathcal{A}_m\rvert)$
MAPPO-PF	$O(\sum_{m\in\mathcal{M}} \lvert\mathcal{S}_m\rvert \cdot \lvert\mathcal{A}_m\rvert)$
MAPPO-SL	$O(\max[(\sum_{m\in\mathcal{M}} \lvert\mathcal{S}_m\rvert \cdot \lvert\mathcal{A}_m\rvert), (\sum_{m\in\mathcal{M}} \lvert\mathcal{A}_m\rvert)^3])$
MAPPO-Lag	$O(\sum_{m\in\mathcal{M}} \lvert\mathcal{S}_m\rvert\lvert\mathcal{A}_m\rvert) + O(M)$

Table VII compares the computational complexity of the algorithms. It is assumed that there are M agents, and each agent's state and action spaces are denoted as $\mathcal{S}_m$ and $\mathcal{A}_m$. And $|\mathcal{S}_m|/|\mathcal{A}_m|$ denotes the size of state/action space. The computational complexity of MASAL involves several steps: The gradient computation has a complexity of $O(\sum_{m\in\mathcal{M}} |\mathcal{S}_m| \cdot |\mathcal{A}_m|)$. The second-order Taylor expansion requires a complexity of $O(M^3)$. And the backtracking line search involves testing different step sizes, adding a complexity of $O(\kappa \sum_{m\in\mathcal{M}} |\mathcal{S}_m| \cdot |\mathcal{A}_m|)$, where κ is the maximum number of line search iterations. Thus, the total complexity of MASAL is approximately $O(M^2 \cdot \sum_{m\in\mathcal{M}} |\mathcal{S}_m| \cdot |\mathcal{A}_m|)$.

Table VIII presents a summary of the computational time for the algorithms investigated, covering a) total number of

TABLE VIII
COMPUTATIONAL TIME COMPARISON

Algorithm	Number of episodes	Total training time (h)	Average test time (ms)
MASAL (proposed)	8,000	26.9	40.2
MAPPO		2.47	21.8
MAPPO-PF		2.47	21.8
MAPPO-SL		2.52	36.4
MAPPO-Lag		8.61	43.9

episodes, b) total training time, and d) average test time, which refers to the computational time needed to determine the joint trading policy on a test day. Although the total training time of the MASAL algorithm is longer than that of other algorithms, their average test times are quite similar.

E. Training Robustness for Changes in Participants' Behavior

1) Role Switching between Buyers and Sellers: In P2P energy trading, participants may alternate between being buyers and sellers depending on their energy production and consumption patterns. For example, the MG may act as a seller during periods of excess solar energy production and as a buyer when its demand exceeds its own generation capacity. As shown in Fig. 11, the proposed framework can accommodate these dynamic role transitions, allowing agents to switch their roles based on real-time energy availability.

2) Trading Frequency: Some participants may engage in the market more frequently, while others may only participate occasionally based on specific needs. The proposed decentralized framework has training robustness for the trading frequency, as the agents who do not participant at a certain time step can bid zero quantity.

3) Bidding Preferences: Participants' bidding preferences include how aggressively or conservatively they set their prices, whether they prefer fixed or flexible pricing, and how they respond to market signals like supply-demand imbalances or price volatility. For instance, participants with a strong risk aversion may prefer stable pricing arrangements, such as setting a personalized range of the bidding price. This can be realized by modifying the action space, e.g., $a^{C}_{1,t} \in [0, 0.8]$ indicating that MG1 is unwilling to bid high prices in the P2P market.

F. Discussion on Line Congestion

The proposed MASAL method is general to incorporate both voltage and line loading constraints. The line capacity constraints can be written as (40)–(42):

$$I_{\mathrm{ij}} = Y_{\mathrm{ij}}\left(V_i - V_j\right) \tag{40}$$

$$S_{\mathrm{ij}} = V_j I^{*}_{\mathrm{ij}} \tag{41}$$

$$-\overline{S}_{\mathrm{ij}} \leq S_{\mathrm{ij}} \leq \overline{S}_{\mathrm{ij}} \tag{42}$$

where I_{ij} denotes the current from node i to node j, I^{*}_{ij} is its complex conjugate, Y_{ij} denotes the complex admittance of the branch ij, S_{ij} represents the apparent power flow from node i to node j and $\overline{S}_{\mathrm{ij}}$ represents the thermal capacity of the branch ij.

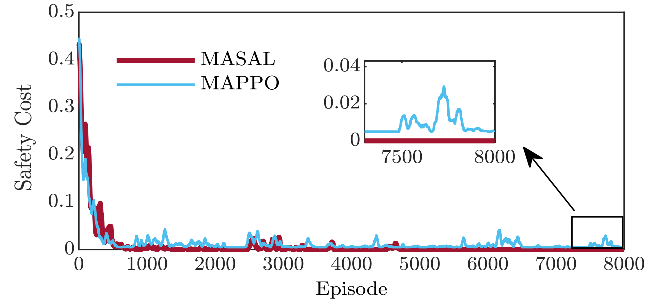

Fig. 14. Training episode cost considering congestion of the primary feeder (scenario 1).

TABLE IX
TRADING OUTCOMES AT 18:00 CONSIDERING LINE CONGESTION OF LINE *1-2*

Algorithm	Seller	Buyer	P2P Quantity (kWh)
MASAL	MG2	MG3	270
MAPPO (unsafe)	MG2	MG3	320
	MG4	MG1	36

Correspondingly, the apparent power flow matrix **S** is added into the observation as follows:

$$o_{m,t} = \left[p^{L}_{m,t}, p^{\mathrm{RES}}_{m,t}, E^{\mathrm{ES}}_{m,t}, \lambda^{\mathrm{FiT}}_{t}, \lambda^{\mathrm{ToU}}_{t}, k^{b}_{t}, k^{s}_{t}, \mathbf{V}, \mathbf{S}\right] \tag{43}$$

And the cost function is designed as the summation of penalties for voltage violations and line congestion as in (44).

$$C_{m,t} = \sum_{i}^{W}\left(\left[V_{i,t} - V^{+}_{i}\right]^{+} + \left[V^{-}_{i} - V_{i,t}\right]^{-}\right) + \sum_{\mathrm{ij}}^{\mathcal{E}}\left(\left[\left|S_{\mathrm{ij}}\right| - \overline{S}_{\mathrm{ij}}\right]^{+}\right) \tag{44}$$

Then, to fully examine the impact of line congestion on P2P transactions, experiments under two different congestion scenarios are conducted.

Scenario 1: Due to the radial structure of the DN, the line connected to the primary node, i.e., line 1-2, experience the highest line loading and are prone to congestion. The capacity of line 1-2 is set at 9.97 kVA. At 18:00, to meet the community net load through import from the upstream grid, line 1-2 is congested. Fig. 14 shows the safety cost considering the congestion of line 1-2 under the proposed MASAL method and the unsafe MAPPO method. Table IX compares the trading outcomes at 18:00 under the proposed MASAL method and the unsafe MAPPO method. It can be observed from Fig. 14 that even under the unsafe MAPPO method, the congestion cost of line 1-2 gradually decreases during training. This is because, in order to increase energy profits (i.e., to raise the reward), the successfully matched P2P trading quantity increases, leading to a reduction in electricity purchases from the upstream network, thus alleviating the congestion on line 1-2. Compared with the unsafe method, the P2P trading quantity is lower under the proposed MASAL method as the MGs reduce their power demand through internal scheduling.

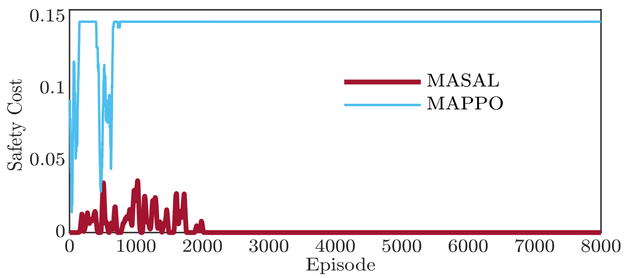

Fig. 15. Training episode cost considering congestion caused by renewable energy injection (scenario 2).

TABLE X
TRADING OUTCOMES AT 21:00 CONSIDERING LINE CONGESTION OF LINE *13-14*

Algorithm	Seller	Buyer	P2P Quantity (kWh)
MASAL	MG2	MG3	92
	MG4	MG1	268
MAPPO (unsafe)	MG2	MG3	270

Scenario 2: In scenarios of high renewable energy penetration, lines of DN are prone to congestion due to reverse power flow. To simulate a high penetration scenario, it is assumed that the power output from the WT at node 14 increases. And the apparent power capacity of line 13-14 is set as 6.3kVA. It turns out that the reverse power flowing from node 14 to node 13 exceeds the line capacity at 21:00. Fig. 15 displays the safety cost considering the congestion caused by the reverse power flow under the proposed MASAL method and the unsafe MAPPO method. It indicates that the participants conduct the responsive trading policies and successfully eliminates the line congestion. To make a specific comparison of how transactions are prioritized, Table X summarizes the trading outcomes at 21:00. Since the power flowing from MG2 (located at node 18) to MG3 (located at node 22) will contribute to the congestion of line 13-14, the energy sold from MG2 to MG3 is reduced by 65.9% under the proposed method compared to the unsafe method. Meanwhile, to reduce energy costs, the transaction of energy sold from MG4 to MG1 is cleared.

G. Discussion on Market Fairness

To maximize the social welfare, MMGs in the community engaged in the P2P trading is cooperative in this study. In this context, the above sections only investigate the community energy costs rather than individual ones. To fairly allocate the energy costs (including the constraint violation costs), a proportional billing allocation method [47] based on the concept of Nash equilibrium can be employed. Equation (45) describes the allocation method.

$$H_m = H_m^{\text{ind}} - \left[\frac{|H_m^{\text{ind}}|}{\sum_{m \in \mathcal{M}} |H_m^{\text{ind}}|} \left(\sum_{m \in \mathcal{M}} H_m^{\text{ind}} - \sum_{m \in \mathcal{M}} H_m \right) \right] \tag{45}$$

where H_m denotes the overall energy costs during the operational day of MG m, H_m^{ind} denotes the individual energy costs of MG m before it joins the community and participates in P2P transactions. The absolute value allows considering MGs with negative energy costs, i.e., energy profits. Equation (45) transforms the collective energy costs of the community into separable individual costs. As can be seen from the previous subsections, to meet the operational constraints, community energy profits may decline. Therefore, the community energy costs already incorporate the costs associated with voltage violations and line congestion, and these costs can be fairly allocated through proportional billing allocation.

V. CONCLUSION AND FUTURE WORK

This paper proposes a novel decentralized multi-agent safe RL method to deal with system-wide global constraint indirectly affected by joint actions. Then, the system operation safety condition classification is proposed to realize safety-aware policy optimization. The method is used to solve network-constrained P2P energy trading problem in RDA market. Case study comparisons verify that 1) The proposed method conducts preventive trading policies and achieves 0 violation under normal conditions during training, better than state-of-the-art safe RL methods. And the obtained reward is 98.64% of that obtained by MAPPO without safety constraint. 2) Under critical conditions, responsive trading policies are conducted. The proposed method makes the safety violation cost monotonically decreased, while the safety layer method and Lagrangian relaxation method fail. 3) The proposed method can generate interpretable bidding policies, scale to larger distribution systems and more P2P participants, and handle changes in participants' behavior and multiple constraints with robustness.

The proposed MASAL method performs well under a static distribution system topology; however, its performance deteriorates when changes in topology occur. Furthermore, high variance in exogenous states can compromise both the economic efficiency and safety guarantees of the system. To address these limitations, our future research will focus on developing a safe MADRL algorithm that is robust to changes in system topology and can handle high variance in exogenous states.

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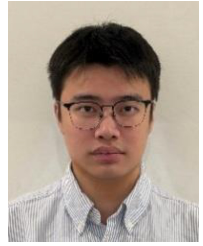

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